

Armor penetration

Lethality redirects here. For the 2013 Mastery, see *Lethality*.

 **Armor penetration** is a champion statistic that determines the amount of a target's  armor that will be ignored when the user deals physical damage. **Percentage armor penetration** and **Lethality** have no effect if the target's armor is less than or equal to 0.

Armor reduction is an effect that reduces the target's armor for a period of time, by an amount or percentage. **Flat** armor reduction can reduce armor values below 0, while **percentage** armor reduction cannot.

It is important to note that **penetration** *treats* the target's resistances as being multiplied by a percentage or reduced by a flat amount, in contrast to **reduction** which applies to the target's resistances as a whole.

All champions have 0 base armor penetration and lethality, but they can be increased by items, runes, and with certain champions' passive ability effects.

Starting in V6.22, **lethality** replaced **flat armor penetration** as a statistic, and offered 40.33% – 100% (based on target's level) of the value as penetration.^[1] In V7.14, it was changed to scale with the user's level instead of the target's level, offering 62.22% – 100% (based on level) of the value as penetration. In V14.1 **lethality** was changed back to no longer scale by level; it grants the full amount of **flat armor penetration** at all levels, but keeps the name to differentiate it from *percent armor penetration* and emphasize its relative effectiveness against low-armor

Contents

Order of calculations

Flat armor reduction

Percentage armor reduction

Percentage armor penetration

Percentage bonus armor penetration

Lethality

Examples

Lethality as scaling

Champion abilities

Items

Increasing lethality

Champion abilities

Items

Item passives

Increasing percent armor penetration

Champion abilities

Items

Item passives

Reducing armor

Champion abilities

Items

Monsters

Notes

Trivia

Highest Lethality

enemies rather than high-armor enemies.^[2] In V14.9, **lethality** was updated to be handled as an innate statistic rather than a scripted one.

Order of calculations

Armor penetration and armor reduction are considered in the following order:^[3]

1. Armor reduction, flat
2. Armor reduction, percentage
3. Armor penetration, percentage
4. Armor penetration, flat (Lethality)

Base armor and **bonus** armor are calculated separately.

Flat armor reduction

The target's armor is reduced by an amount. Flat armor reduction + stacks additively and is distributed proportionally between base armor and bonus armor.

Example: 15 armor reduction against a 20 base armor and 40 bonus armor target will reduce base armor by 5 and bonus armor by 10, so the target will be reduced to 45 armor $((20 - 5) + (40 - 10) = 45)$.

Flat armor reduction can reduce a target's armor below zero. For example, if an enemy with 10 armor has their armor reduced by 25, the enemy will have −15 armor.

Percentage armor reduction

The target's armor is multiplied by a percentage (100% – the listed reduction). Applies to both base and bonus armor. Percentage armor reduction × stacks multiplicatively and is ignored if the target's armor is 0 or less.

Example: 30% armor reduction multiplies the target's armor to 70%, so a target with 20 base armor and 40 bonus armor will be reduced to 42 armor $((20 \times 0.7) + (40 \times 0.7) = 42)$.

Percentage armor reduction reduces more armor on targets with higher armor. For instance, with 40% armor reduction, a target with 200 armor will lose 80 while a target with only 50 armor will lose 20.

With 40% armor reduction, the physical damage dealt to a target with 50 armor is only increased by around 15%, whereas the physical damage dealt to a target with 200 armor is increased by around 36%.

If one wanted to calculate the total armor reduction after two consecutive percent armor reductions, the following formula may be useful:

$$\text{CumulativePercentArmorReduction} = r_1 + r_2 - r_1 r_2$$

Highest Armor Penetration (%)

Media

See also

References

Each r is the individual percent armor reductions. For example, the calculation of two armor reductions, 20% and 30%, would yield the following:

$$\text{CumulativePercentArmorReduction} = r_1 + r_2 - r_1 r_2 = (.2) + (.3) - (.2)(.3) = 44\%$$

Percentage armor penetration

The target's armor is treated as being multiplied by a percentage for purposes of damage calculation (100% – the listed penetration). Percentage armor penetration  stacks multiplicatively and is ignored if the target's armor is 0 or less.

Example: 30% armor penetration causes the target to take damage as if its armor was multiplied by 70%, so a target with 20 base armor and 40 bonus armor will be treated as though it had 42 for purposes of damage (the target's actual armor value will not change, however) $((20 \times 0.7) + (40 \times 0.7) = 42)$.

Because the target's actual armor value does not change, anything else depending on that value also doesn't change.

Percentage armor penetration reduces more armor on targets with higher armor. For instance, with 40% armor penetration, a target with 200 armor will be considered as having 80 less while a target with only 50 armor will be considered as having 20 less.

Gold Value

Percent armor penetration has a gold value of  41.66667 per point.

Percentage bonus armor penetration

Percentage armor penetration that applies only to **bonus** armor.

Example: 30% bonus armor penetration multiplies the target's bonus armor to 70% without affecting base armor, so a target with 20 base armor and 40 bonus armor will be treated as though it had 48 for purposes of damage calculation (the target's actual armor value will not change, however). $(20 + (40 \times 0.7) = 48)$.

If the unit has multiple sources of percentage bonus penetration, they stack multiplicatively:

Example: 10% armor penetration and 30% bonus armor penetration. $(20 \times 0.9) + (40 \times 0.7 \times 0.9) = 43.2$.

Because the target's actual armor value does not change, anything else depending on that value also doesn't change.

Percentage bonus armor penetration reduces more armor on targets with higher **bonus** armor, but will not have an effect against base armor. For instance, with 40% **bonus** armor penetration, a target with 200 **bonus** armor will be considered as having 80 less while a target with only 50 **bonus** armor will be considered as having 20 less. In contrast, a target with any amount of base armor and no **bonus** armor will be considered as having the same value.

Lethality

Lethality is the statistic that represents and grants flat armor penetration.

The target's armor is treated as being reduced by an amount for the purposes of damage calculation, but cannot be reduced below 0. Flat armor penetration  stacks additively.

Example: 20 lethality on a user will cause a target to take damage as if its armor was 20 less, so a target with 30 armor will be treated as though it had 10 for purposes of damage calculation (the target's actual armor value will not change, however).

Because the target's actual armor value does not change, anything else depending on that value also doesn't change.

Gold Value

Lethality has a gold value of  30 per point, for more details see [this page](#).

Examples

Given 10 flat armor penetration and 45% bonus armor penetration, and the target is affected by 30 flat armor reduction and 30% armor reduction;

Target A has 300 armor (100 base and 200 bonus armor).

The 300 is reduced to 270 (90 base and 180 bonus armor) by the 30 armor reduction.

The 270 is reduced to 189 (63 base and 126 bonus armor) by the 30% armor reduction.

The 189 is considered to be 132.3 (63 base and 69.3 bonus armor) by the 45% bonus armor penetration.

The 132.3 is considered to be 122.3 by the 10 armor penetration.

Target A takes damage as if it has 122.3 armor.

Target B has 18 armor.

The 18 is reduced to -12 by the 30 armor reduction.

The -12 is not affected by any further calculations because it is less than 0.

Target B takes damage as if it has -12 armor.

Lethality as scaling

The following have their stats improved based on the amount of lethality acquired:

Champion abilities

 [Pyke's Undertow](#),  [Gift of the Drowned Ones](#),  [Ghostwater Dive](#),  [Phantom Death from Below](#)

Items

 [Axiom Arc](#)

Increasing lethality

Champion abilities



Aphelios'



The Hitman and the Seer

Items

This table is automatically generated based on the data from Module:ItemData/data.

Item	Cost	Amount	Availability
 Axiom Arc	3000 	18	Nexus Blitz, Arena, Classic Summoner's Rift 5v5, All Random All Mid
 Duskblade of Draktharr	0 	20	Arena
 Edge of Night	3000 	15	Nexus Blitz, Arena, Classic Summoner's Rift 5v5, All Random All Mid
 Guardian's Dirk	500 	11	Arena
 Hellfire Hatchet	2500 	12	Arena
 Hubris	3000 	18	Nexus Blitz, Arena, Classic Summoner's Rift 5v5, All Random All Mid
 Opportunity	2700 	15	Nexus Blitz, Arena, Classic Summoner's Rift 5v5, All Random All Mid
 Profane Hydra	3200 	18	Nexus Blitz, Arena, Classic Summoner's Rift 5v5, All Random All Mid
 Prowler's Claw	0 	22	Arena
 Regicide	0 	15	Arena
 Serpent's Fang	2500 	15	Nexus Blitz, Arena, Classic Summoner's Rift 5v5, All Random All Mid
 Serrated Dirk	1000 	10	Nexus Blitz, Classic Summoner's Rift 5v5, All Random All Mid
 Spectral Cutlass	2500 	21	Arena
 Talisman of Ascension	0 	0	Arena
 The Brutalizer	1337 	5	Nexus Blitz, Classic Summoner's Rift 5v5, All Random All Mid
 The Collector	3000 	10	Nexus Blitz, Arena, Classic Summoner's Rift 5v5, All Random All Mid
 Umbral Glaive	2500 	15	Nexus Blitz, Classic Summoner's Rift 5v5, All Random All Mid
 Voltaic Cyclosword	3000 	18	Nexus Blitz, Arena, Classic Summoner's Rift 5v5, All Random All Mid
 Youmuu's Ghostblade	2800 	18	Nexus Blitz, Arena, Classic Summoner's Rift 5v5, All Random All Mid

Item passives

 [Opportunity](#)

Increasing percent armor penetration

Champion abilities

Ambessa's Public Execution
 Darius' Apprehend

K'Sante's All Out
 Nilah's Formless Blade
 Pantheon's Grand Starfall

Items

This table is automatically generated based on the data from [Module:ItemData/data](#).

Item	Cost	Amount	Availability
Last Whisper	1450	18%	Nexus Blitz, Classic Summoner's Rift 5v5, All Random All Mid
Lord Dominik's Regards	3100	40%	Nexus Blitz, Arena, Classic Summoner's Rift 5v5, All Random All Mid
Mortal Reminder	3300	35%	Nexus Blitz, Arena, Classic Summoner's Rift 5v5, All Random All Mid
Perplexity	2500	22%	Arena
Serylda's Grudge	3000	35%	Nexus Blitz, Arena, Classic Summoner's Rift 5v5, All Random All Mid
Talisman of Ascension	0	0%	Arena

Item passives

Terminus

Reducing armor

Champion abilities

Percentage Reduction

Briar's Head Rush
 Garen's Judgment
 Jarvan IV's Dragon Strike
 Jayce's Transform Mercury Cannon
 Kayle's Radiant Blast
 Kog'Maw's Caustic Spittle
 Mordekaiser's Realm of Death

Nasus' Spirit Fire
 Olaf's Undertow
 Rell's Break the Mold
 Renekton's Dice (Empowered)
 Sion's Roar of the Slayer
 Trundle's Subjugate
 Vi's Denting Blows
 Wukong's Crushing Blow

Flat Reduction

-  [Corki's](#)  [Gatling Gun](#)
-  [Rell's](#)  [Break the Mold](#)
-  [Rengar's](#)  [Thrill of the Hunt](#)

Items

Percentage Reduction

-  [Black Cleaver](#)

Monsters

Flat Reduction

-  [Atakhan's](#)  [Frail and Unworthy](#)
-  [Baron Nashor's](#)  [Void Corruption](#)

Notes

Armor penetration will apply to any and all sources of  [physical damage](#).

It will apply to the damage dealt to almost every type of unit, including  [turrets](#) but not other structures ([Inhibitors](#) are the only building with armor).

Armor reduction will affect the physical damage from any unit source.

Percentage penetration is applied before flat penetration.

 [Turrets](#) and  [Dragons](#) have 30% armor penetration.

 [Gangplank's](#)  [Powder Keg](#) is special cased to ignore a percentage of the target's armor with its damage.

 [Yasuo's](#)  [Last Breath](#) grants him a buff that causes the damage dealt by his critical strikes to ignore a percentage of the target's bonus armor.

Trivia

Highest Lethality

Last updated: January 20, 2024, patch [V14.10](#)

The highest amount of lethality any champion can obtain is **127**.

Items:

-  [Certainty](#) - 27

This item grants an additional 10 lethality at the highest level with its passive effect, [Preparation](#).

-  [Yuumu's Ghostblade](#) - 18

 Voltaic Cyclosword - 18

 Hubris - 18

 Axiom Arc - 18

 Profane Hydra - 18

Calculation:

Items =  27 + 10 +  18 +  18 +  18 +  18 +  18 = **127 lethality**

The highest amount of lethality obtainable in the game is **160** with  Aphelios when the above items are equipped and  The Hitman and the Seer's Lethality is at rank 5.

127 +  33's = **160**

Highest Armor Penetration (%)

The highest amount of percentage armor penetration any champion can obtain is **40%** with  Lord Dominik's Regards.

The highest amount of percentage armor penetration obtainable in the game is **64%** with  Darius when  Lord Dominik's Regards is equipped and  Apprehend is at rank 5.

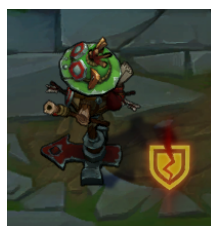
Use the formula for armor:

$$(1 - \text{img alt="Lord Dominik's Regards icon" data-bbox="335 506 365 526"} 0.4) \times (1 - \text{img alt="Lord Dominik's Regards icon" data-bbox="445 506 475 526"} 0.4) = 0.36$$

Convert to armor penetration:

$$(1 - 0.36) \times 100\% = \mathbf{64\% \text{ armor penetration}}$$

Media



Armor
indicator



reduced Armor and magic
resistance reduced
indicator

See also

Armor

Magic penetration

Champion statistic

References

1. <https://web.archive.org/web/0/http://na.leagueoflegends.com/en/news/game-updates/patch/patch-72-notes#patch-items>
2. Lethality name (<https://youtu.be/PY4boPCTeoo?t=10745>)
3. If I technically had 100% armor reduction on an enemy... (<https://web.archive.org/web/0/http://na.leagueoflegends.com/board/showthread.php?p=32707861#32707861>) at LeagueOfLegends.com Boards

Champion statistics

[show]

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